SGN – Assignment #1

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1 Lambert problem

Exercise 1

Consider the 3D Earth–Moon Circular Restricted Three-Body Problem (CRTBP) with $\mu = 0.012150$. The Lambert problem, which is stated as the search of a path between two given points within a specified time of flight, is a well-known problem in the two-body model. However, its solution within the CRTBP requires leveraging numerical techniques.

1) Solve the Lambert problem in the CRTBP for the given position vectors $\boldsymbol{r}_1 = [-0.3; 0; 0]$ and $\boldsymbol{r}_2 = [0.008; -0.01; 0]$ with a time of flight $\Delta t = 0.9$ (all parameters are adimensional). To do so, develop the theoretical framework based on the expansion of the flow and implement a differential correction scheme that uses the STM, which can be either approximated through finite differences **or** achieved by integrating the variational equation. Consider a first guess for the initial velocity $\boldsymbol{v}_1^g = [0; 1; 0]$. Plot the final solution and report its numerical values in Table 1.

A one-dimensional family of solutions can be found by varying the time of flight. *Numerical continuation* is an iterative process in which a variable is *gradually* varied until the desired value is reached, while the initial guess is taken from the solution of the previous iteration. This procedure facilitates the convergence process of the numerical solver.

2) By gradually decreasing the time of flight, and using numerical continuation, compute the families of paths connecting $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ until $\Delta t = 0.3$. Plot the resulting intermediate solutions and report the numerical values of the final solution in Table 3.

The final position $\boldsymbol{r}_2 = [0.008; -0.01; 0]$ (the same as Point 1) belongs to a Low Earth Orbit (LEO). Once a connecting path between $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ is found, the goal is to perform an impulsive maneuver to inject the spacecraft into the target LEO. Consider the adimensional LEO target velocity $\boldsymbol{v}_{\text{LEO}} = [-2.56; -5.16; 3.31]$.

3) By gradually decreasing Δt, and using numerical continuation, find the time of flight $\Delta t_{\min}$ that produces the minimum injection maneuver $\Delta v_{\min}$. Plot the trend of the injection maneuver Δv as a function of the time of flight Δt and report the numerical values of $\Delta t_{\min}$ and $\Delta v_{\min}$ in Table 6.

(8 points)

1.1 Lambert Arc Solution for $\Delta t = 0.9$

- **Objective** To solve the Lambert problem in the Circular Restricted Three-Body Problem (CR3BP) for the given position vectors $\mathbf{r}_1 = [-0.3; 0; 0]$ and $\mathbf{r}_2 = [0.008; -0.01; 0]$ with a time of flight $\Delta t = 0.9$ (adimensional). The solution requires implementing a differential correction scheme based on the State Transition Matrix (STM) to determine the initial velocity $\mathbf{v}_1$ that connects the two positions in the specified time.

- **Methodology & Code Implementation** A single-shooting Newton corrector is implemented in the CR3BP synodic frame to solve the Lambert boundary-value problem. The implementation follows these principles:

- **Problem Statement:** Given fixed initial position $\mathbf{r}_1$, target position $\mathbf{r}_2$, and time-of-flight Δt, determine the initial velocity $\mathbf{v}_1$ such that the trajectory arrives precisely at $\mathbf{r}_2$ after time Δt. The shooting function quantifies the terminal position mismatch:

$$\mathbf{F}(\mathbf{v}_1) = \mathbf{r}_2 - \mathbf{r}(t_f;\, \mathbf{r}_1, \mathbf{v}_1) \tag{1}$$

- **Theoretical Framework:** The correction strategy relies on linearizing the flow map via first-order Taylor expansion. A small perturbation in the initial state $\delta\mathbf{x}_0$ induces a variation at the final time governed by:

$$\delta\mathbf{x}(t_f) \simeq \mathbf{\Phi}(t_f, t_0)\, \delta\mathbf{x}_0 \tag{2}$$

where $\mathbf{\Phi}(t_f, t_0) \in \mathbb{R}^{6\times 6}$ is the State Transition Matrix encoding trajectory sensitivity to initial conditions.

- **STM Propagation:** The STM is computed by integrating the variational equations simultaneously with the CR3BP dynamics:

$$\dot{\mathbf{\Phi}}(t, t_0) = \mathbf{A}(t)\, \mathbf{\Phi}(t, t_0), \qquad \mathbf{\Phi}(t_0, t_0) = \mathbf{I}_{6\times 6} \tag{3}$$

where $\mathbf{A}(t) = \partial\mathbf{f}/\partial\mathbf{x}$ is the Jacobian of the CR3BP vector field evaluated along the propagated trajectory. Computing the STM via variational equations requires only one integration of the augmented system, eliminating the repeated trajectory propagations that finite-difference column-by-column approximations would demand and avoiding the associated numerical degradation.

- **Constraint Application and System Reduction:** For the Lambert problem, the initial position is fixed ($\Delta\mathbf{r}_0 = \mathbf{0}$) and the final velocity remains free. Partitioning the STM into 3×3 blocks and applying these constraints reduces the full 6×6 sensitivity relation to:

$$\delta\mathbf{r}_f \simeq \mathbf{\Phi}_{rv}(t_f, t_0)\, \Delta\mathbf{v}_0 \tag{4}$$

where $\mathbf{\Phi}_{rv} \in \mathbb{R}^{3\times 3}$ maps initial velocity variations into final position variations. Identifying $\delta\mathbf{r}_f = \mathbf{F}(\mathbf{v}_1)$, the velocity correction is obtained by inverting this relationship.

- **Newton-Raphson Iteration:** At each iteration k, the correction is computed and applied:

$$\Delta\mathbf{v}_1^{(k)} = \mathbf{\Phi}_{rv}^{-1}(t_f)\, \mathbf{F}(\mathbf{v}_1^{(k)}), \qquad \mathbf{v}_1^{(k+1)} = \mathbf{v}_1^{(k)} + \Delta\mathbf{v}_1^{(k)} \tag{5}$$

- **Numerical Integration:** The augmented state-STM system (42 ODEs: 6 for state, 36 for STM elements) was integrated using `ode113` The Earth-Moon mass parameter is $\mu = 0.012150$.

- **Initial Guess and Convergence:** An initial velocity guess $\mathbf{v}_1^{\text{guess}} = [0; 1; 0]^T$ initiates the iteration. Convergence is declared when $\|\mathbf{F}\|$ falls below the prescribed tolerance or the maximum iteration count is reached.

- **Results & Analysis** The differential corrector converged successfully in 15 iterations. The dimensional conversion factors used throughout this exercise are derived from the Earth-Moon system characteristics:

$$\text{DU} = 384400 \text{ km}, \quad \text{TU} = \sqrt{\frac{\text{DU}^3}{\mu_{\text{EM}}}} = 4.342 \text{ days}, \quad \text{VU} = \frac{\text{DU}}{\text{TU}} = 1.025 \text{ km/s} \tag{6}$$

where DU is the Earth-Moon distance, $\mu_{\text{EM}} = G(M_\oplus + M_{\text{Moon}})$ is the combined gravitational parameter of the Earth-Moon system, TU is the characteristic orbital period scale, and VU is the velocity scale. The corrected initial and terminal velocities at $\Delta t = 0.9$ are reported in Table 1.

Table 1: Corrected initial velocity at $\Delta t = 0.9$ [-]

v_x [-]	v_y [-]	v_z [-]
-1.7424051346	$+0.9576314537$	$+0.0000000000$

Table 2: Terminal velocity $\mathbf{v}_f(t_f)$ at $\Delta t = 0.9$ [-]

$v_x(t_f)$ [-]	$v_y(t_f)$ [-]	$v_z(t_f)$ [-]
-6.9299742377	-6.0670887307	$+0.0000000000$

The resulting Lambert arc is shown in Figure 1, which depicts the trajectory connecting $\mathbf{r}_1$ to $\mathbf{r}_2$ in the CR3BP synodic reference frame. The arc exhibits the characteristic curved shape influenced by the gravitational fields of both primaries. The CR3BP model is inherently three-dimensional, but because the terminal position constraint imposes $z_2 = 0$, the initial position satisfies $z_1 = 0$, and the initial velocity guess lies in the xy-plane ($\mathbf{v}_1^{\text{guess}} = [0; 1; 0]^T$), the differential corrector evolves entirely within the xy planar where $z = 0$ and $v_z = 0$ throughout all iterations consistent with the zero v_z entries in Tables 1 and 2.

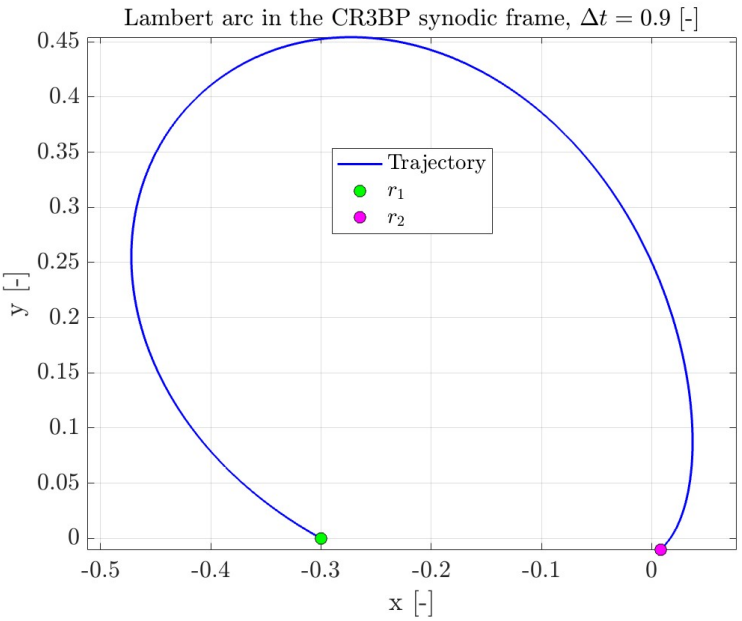


Figure 1: Lambert arc in the CR3BP synodic frame for $\Delta t = 0.9$ [-].

1.2 Numerical Continuation (Δt: 0.9 to 0.3)

- **Objective** To compute a one-dimensional family of Lambert solutions by gradually decreasing the time of flight from $\Delta t = 0.9$ to $\Delta t = 0.3$ using numerical continuation.
- **Methodology & Code Implementation** Numerical continuation is an iterative process in which a parameter (in this case, Δt) is gradually varied, and the solution from the previous parameter value serves as the initial guess for the current iteration. The procedure was implemented as follows:

  1. **Parameter Discretization:** The time-of-flight range was discretized into equally-spaced points from $\Delta t = 0.9$ to $\Delta t = 0.3$.
  2. **Warm-Start Strategy:** For each Δt_i, the differential corrector was initialized with

the converged velocity from the previous step $\mathbf{v}_1(\Delta t_{i-1})$, except for the first step which used the Part 1 solution.

3. **Corrector Iteration:** At each parameter value Δt_i, the Newton-Raphson algorithm (identical to Part 1) was applied until convergence.

- **Results & Analysis** The continuation process converged successfully for all parameter values of Δt . The terminal solution at $\Delta t = 0.3$ is reported in Table 3, confirming that the final position matches the target $\mathbf{r}_2$ to within the prescribed tolerance.

Table 3: Corrected initial velocity at $\Delta t = 0.3$ [-]

v_x [-]	v_y [-]	v_z [-]
−0.7250227523	+0.9959942636	+0.0000000000

Table 4: Terminal velocity $\mathbf{v}_f(t_f)$ at $\Delta t = 0.3$ [-]

$v_x(t_f)$ [-]	$v_y(t_f)$ [-]	$v_z(t_f)$ [-]
−4.0849007020	−8.1063138211	+0.0000000000

Table 5: Terminal position $\mathbf{r}_f(t_f)$ at $\Delta t = 0.3$ [-]

r_x [-]	r_y [-]	r_z [-]
+0.0080000000	−0.0100000000	+0.0000000000

Terminal position error: $\|\mathbf{r}_f - \mathbf{r}_2\| = 4.310 \times 10^{-15}$ [-]

Figure 2 shows the complete family of Lambert arcs obtained through continuation.

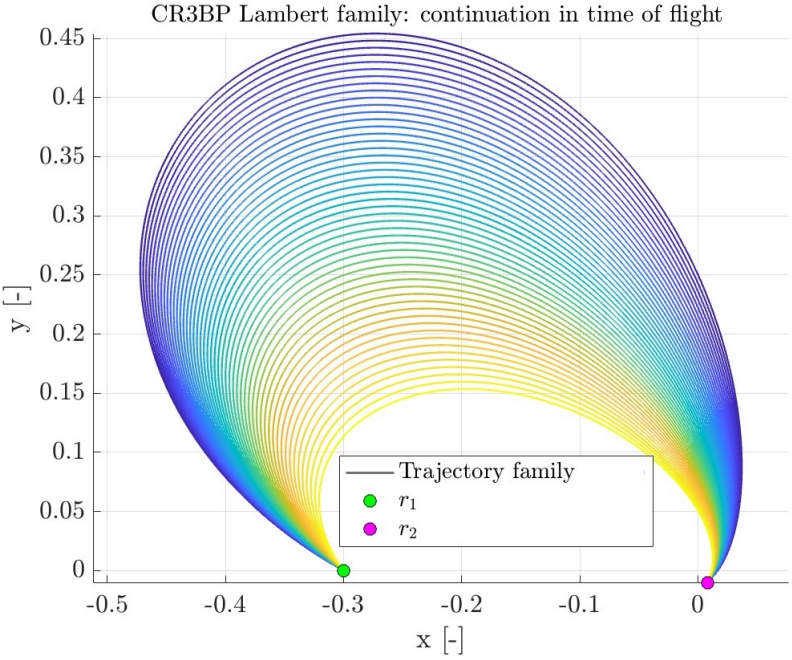


Figure 2: CR3BP Lambert family obtained through continuation in time of flight. The color gradient indicates decreasing time of flight from dark blue (long TOF) to yellow (short TOF).

1.3 Minimum Injection Maneuver Optimization

- **Objective** The goal is to find $\Delta t_{\min}$ minimizing $\Delta v = \|\mathbf{v}_{\text{LEO}} - \mathbf{v}_f(\Delta t)\|$, with $\mathbf{v}_{\text{LEO}} = [-2.56;\ -5.16;\ 3.31]$, via a hierarchical grid search with geometrically shrinking brackets.

- **Methodology & Code Implementation** The optimization was performed using a multi-level grid search with progressive geometric refinement:

  1. **Cost Function Definition:** For a given time of flight Δt, the Lambert problem is solved via differential correction to obtain the terminal velocity $\mathbf{v}_f(\Delta t)$. The injection cost is then:
  $$J(\Delta t) = \|\mathbf{v}_{\text{LEO}} - \mathbf{v}_f(\Delta t)\| \tag{7}$$
  2. **Initial Coarse Scan:** A coarse grid of 750 equally-spaced points spanning $\Delta t \in [0.300, 0.001]$ was evaluated. At each grid point, the differential corrector was applied with warm-starting from the previous converged solution, and the cost $J(\Delta t)$ was computed.
  3. **Geometric Bracketing Strategy:** After each level, the minimum Δt^* on the current grid is identified. The bracket for the next zoom pass is defined geometrically as:
  $$[\alpha_{\text{lo}},\ \alpha_{\text{hi}}] = [\max(0.001,\ \Delta t^* - \delta),\ \min(0.300,\ \Delta t^* + \delta)] \tag{8}$$
  where $\delta = \eta \cdot |\Delta t_{\text{end}} - \Delta t_{\text{start}}|/2$ and $\eta = 0.20$ is the shrink factor, concentrating the search interval around the minimum at each level.
  4. **Refinement & Convergence:** Five zoom passes (250 pts each) are executed within geometrically updated brackets, warm-started from the converged velocity nearest to α_{lo} for jump-free continuation, and terminated when $|\alpha_{\text{hi}} - \alpha_{\text{lo}}| < \varepsilon_{\text{bracket}}$ or all passes are completed. A good solution can be reached with 3 zoom passes already.

  All evaluated $(\Delta t, \Delta v)$ pairs across all levels were accumulated for comprehensive visualization of the cost landscape.

- **Results & Analysis** The hierarchical optimization successfully localized the minimum injection maneuver after 1 coarse sweep and 5 zoom passes. The optimal parameters are reported in Table 6 in both adimensional and dimensional (SI) units obtained via:
$$\Delta t\ [\text{days}] = \Delta t\ [-] \cdot \text{TU}_{\text{days}}, \qquad \Delta v\ [\text{km/s}] = \Delta v\ [-] \cdot \text{VU}. \tag{9}$$

Table 6: Time of flight and corresponding impulse magnitude for minimum injection maneuver

$\Delta t_{\min}$ [-]	$\Delta v_{\min}$ [-]	$\Delta t_{\min}$ [days]	$\Delta v_{\min}$ [km/s]
0.272675	4.681657	1.184088	4.796577

Figure 3 shows the complete cost landscape: the function decreases for large Δt, reaches its minimum at $\Delta t_{\min} \approx 1.184$ days (blue cross), then rises sharply for short transfers.

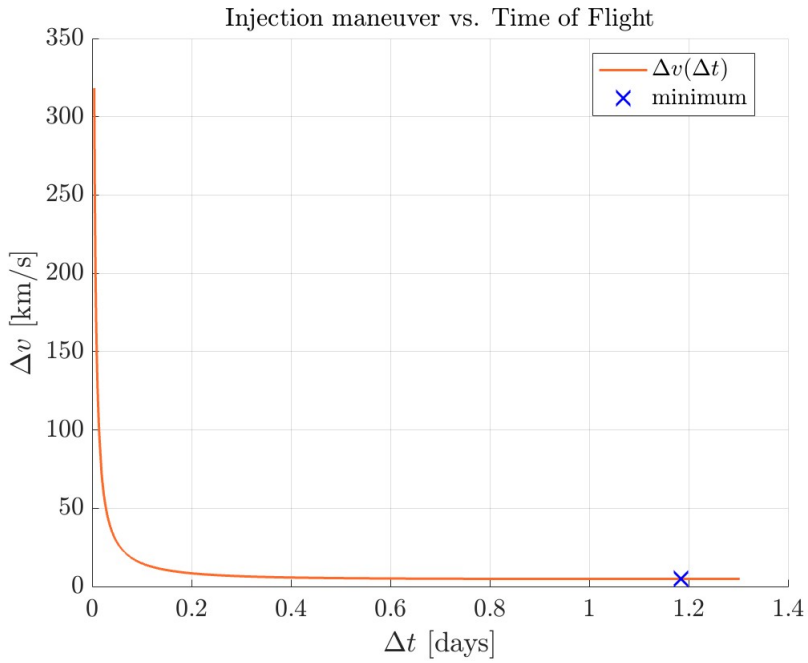


Figure 3: Injection Δv versus time of flight Δt over full search domain showing 6-level hierarchical refinement (1 coarse + 5 zoom passes).

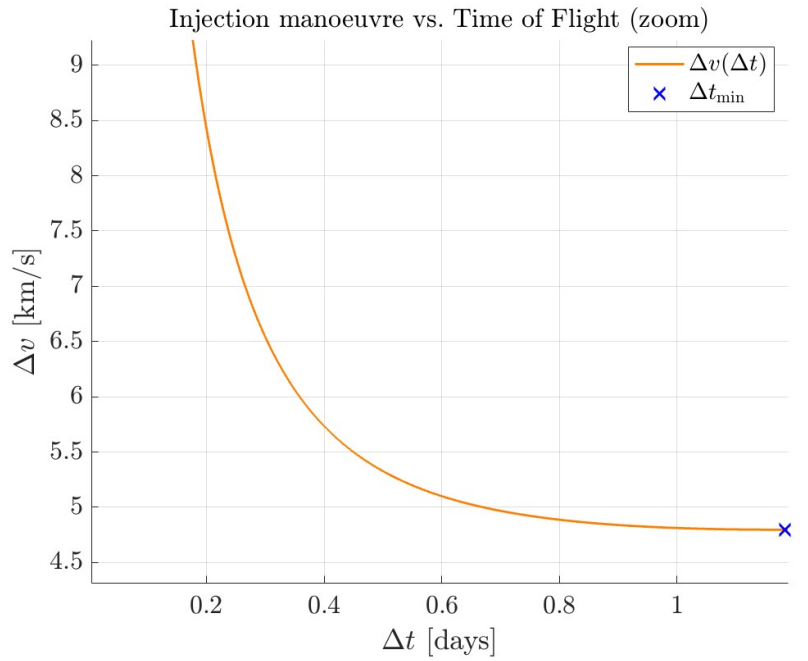


Figure 4: Zoomed view near $\Delta t_{\min}$.

2 Continuous guidance

Exercise 2

A CubeSat is performing operations around the astroid Eros. Due to very small gravity, the spacecraft is equipped with a very-low-thrust electric propulsion system. The spacecraft shall perform an orbit lowering maneuver, starting from its parking orbit at 52.25 km altitude and descending till a circular orbit at 36.15 km altitude above the asteroid surface. Both orbits are on the ecliptic plane. The observations performed by the CubeSat reported the presence of dust and small particles around the asteroid, whose spatial density q as function of the distance from the asteroid center ρ (in km) has been determined as follows:

$$q(\rho) = \frac{k_1}{k_2 + \left(\frac{\rho - \rho_A}{DU}\right)^2} + \frac{k_3}{k_4 + \left(\frac{\rho - \rho_B}{DU}\right)^2}$$

where ρ is the distance from the Eros center. The objective is to design an optimal orbit lowering that minimizes interaction with the dust in order to preserve the solar panels. This corresponds to minimizing the following objective function:

$$F(t) = \int_{t_0}^{t_f} q(\rho(t))\,\mathrm{d}t.$$

The parameters and reference Distance and Mass Unit to be considered are given in Table 7.

Symbol	Value	Units	Meaning
h_i	52.25	km	Altitude of departure orbit
h_f	36.15	km	Altitude of arrival orbit
R_a	17.00	km	Eros radius
μ	3.5×10^{-4}	$\mathrm{km}^3/\mathrm{s}^2$	Eros gravitational parameter
ρ_A	$23.314 + R_a$	km	Radius of first dust density peak
ρ_B	$42.170 + R_a$	km	Radius of second dust density peak
k_1	7.393750×10^{-3}	DU^{-1}	Dust spatial density constant 1
k_2	7.500000×10^{-3}	DU^2	Dust spatial density constant 2
k_3	3.696875×10^{-4}	DU^{-1}	Dust spatial density constant 3
k_4	6.250000×10^{-4}	DU^2	Dust spatial density constant 4
m_0	25.0	kg	Initial mass
T	21.59	μN	Maximum thrust
I_{sp}	382.82	s	Specific impulse
DU	69.25	km	Distance Unit
MU	m_0	kg	Mass Unit

Table 7: Problem parameters and constants. The units of time TU and velocity VU can be computed imposing that the scaled gravitational parameter $\overline{\mu} = 1$.

1) Plot the debris spatial density $q(\rho) \in [20; 70]$ km and compute the initial state and target orbital state, knowing that: i) the initial state is on the circular orbit at altitude h_i with an angle of +45 deg with respect to the x-axis of the ecliptic J2000 reference frame; ii) the final state is located on the x-axis of the ecliptic J2000 reference frame; **Hint**: the two orbits are coplanar.

2) Adimensionalize the problem using as reference length $DU = \rho_i = h_i + R_a$ and reference mass $MU = m_0$, imposing that $\mu = 1$. Report all adimensionalized parameters.

3) Using the PMP, write down the spacecraft equations of motion, costate dynamics, and the zero-finding problem for unknowns $\{\boldsymbol{\lambda}_0, t_f\}$ with the appropriate transversality condition. **Hint**: the spacecraft has to reach the target state computed in point 1).

4) Solve the problem considering the data provided in Table 7. To obtain an initial guess for the costate, generate random numbers such that $\lambda_{0,i} \in [-100; +100]$ and considering less than three revolutions to complete the transfer. Report the solution obtained in terms of $\{\boldsymbol{\lambda}_0, t_f\}$ and the error with respect to the target. Assess your results by exploiting the properties of the Hamiltonian in problems that are not time-dependent and time-optimal solutions.

5) To perform better observations of the asteroid surface, you are asked to reach a slightly different orbit with an inclination $\Delta i = 3.5$ deg above the ecliptic. Assume that the rotation of the rotation of the angle Δi is performed around the x-axis of the ecliptic J2000 reference frame (RAAN = 0). Solve the problem and compare the new solution with the one obtained in the previous point. **Hint**: exploit numerical continuation.

6) Perform an n-body propagation using the final state obtained in Point 5 as the initial condition for the numerical integration, taking Eros as the central body. Consider a propagation interval from the initial epoch `2012 Jan 15 00:00:00.000 TDB` to the final epoch `2012 Feb 15 00:00:00.000 TDB`. Plot the resulting trajectory and compare it with the corresponding Keplerian solution.

(11 points)

2.1 Debris Environment Analysis and Boundary State Computation

- **Objective**
  - To characterize the debris density environment around asteroid Eros and establish the dimensional boundary states for the optimal low-thrust transfer. The debris density model $q(\rho)$ is analyzed to identify peak hazard zones, and the initial and target state vectors are computed in the Eros-centered ecliptic J2000 inertial reference frame.
- **Methodology & Code Implementation**
  - **Debris Density Model.** The debris spatial density is modeled as a sum of two Lorentzian-shaped peaks:

$$q(\rho) = \frac{k_1}{k_2 + \left(\dfrac{\rho - \rho_A}{DU}\right)^2} + \frac{k_3}{k_4 + \left(\dfrac{\rho - \rho_B}{DU}\right)^2} \tag{10}$$

    where ρ is the radial distance from the Eros center in km, $\rho_A = 23.314 + R_a$ km and $\rho_B = 42.170 + R_a$ km are the nominal peak centers, and $DU = 69.25$ km is the reference distance unit. The density peak locations were identified numerically by scanning $q(\rho)$ over two separate intervals: $\rho \in [20, 50]$ km for Peak A and $\rho \in [50, 70]$ km for Peak B, using a fine-grained linspace discretization followed by a `max` search.
  - **Orbital Radii and Circular Velocities.** The radii of the departure and arrival circular orbits are computed directly from the given altitudes:

$$r_i = R_a + h_i = 69.25 \text{ km}, \qquad r_f = R_a + h_f = 53.15 \text{ km} \tag{11}$$

    The corresponding circular orbital speeds follow from the vis-viva relation for circular motion:

$$v_i = \sqrt{\mu/r_i}, \qquad v_f = \sqrt{\mu/r_f} \tag{12}$$

– **Initial State Construction.** Both orbits lie in the ecliptic plane ($z = 0$). The initial state is placed at $\theta_i = +45$ from the x-axis of the ecliptic J2000 frame. The position vector follows from standard polar-to-Cartesian conversion:

$$\mathbf{r}_0 = r_i \begin{bmatrix} \cos\theta_i \\ \sin\theta_i \\ 0 \end{bmatrix} \tag{13}$$

For a prograde circular orbit, the velocity vector is perpendicular to $\mathbf{r}_0$ and obtained by applying a $+90$ rotation about the z-axis to the position unit vector:

$$R_z(90) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \implies \mathbf{v}_0 = v_i \begin{bmatrix} -\sin\theta_i \\ \cos\theta_i \\ 0 \end{bmatrix} \tag{14}$$

– **Target State Construction.** The final state is located on the positive x-axis of the ecliptic J2000 frame ($\theta_f = 0$). The position and velocity vectors reduce to:

$$\mathbf{r}_f = \begin{bmatrix} r_f \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_f = \begin{bmatrix} 0 \\ v_f \\ 0 \end{bmatrix} \tag{15}$$

where the velocity points purely in the $+y$ direction, consistent with a prograde circular orbit crossing the x-axis.

- **Results & Analysis**

– **Debris Density Profile:** The numerical peak search yields the locations reported in Table 8, while Figure 5 illustrates the spatial density $q(\rho)$ over the range [20, 70] km. The inner peak at $\rho_A \approx 40.32$ km corresponds to the inner debris belt and reaches a maximum density of approximately 1.0 DU^{-1}. The outer peak at $\rho_B \approx 59.15$ km defines the outer belt and attains roughly 0.68 DU^{-1}.

Table 8: Debris density peak locations identified by numerical maximization of $q(\rho)$.

Metric	**Peak A**	**Peak B**
Distance [km]	40.3236	59.1478
Distance [DU]	0.5823	0.8541

A local minimum between the two peaks defines a comparatively safer corridor. Both peaks lie within the transfer corridor between the departure radius $r_i = 69.25$ km and the arrival radius $r_f = 53.15$ km. Specifically, the transfer corridor $[r_f, r_i]$ fully encompasses Peak B and borders Peak A, confirming that the optimal trajectory must actively negotiate these hazard zones and motivating the optimal control formulation in the subsequent sections.

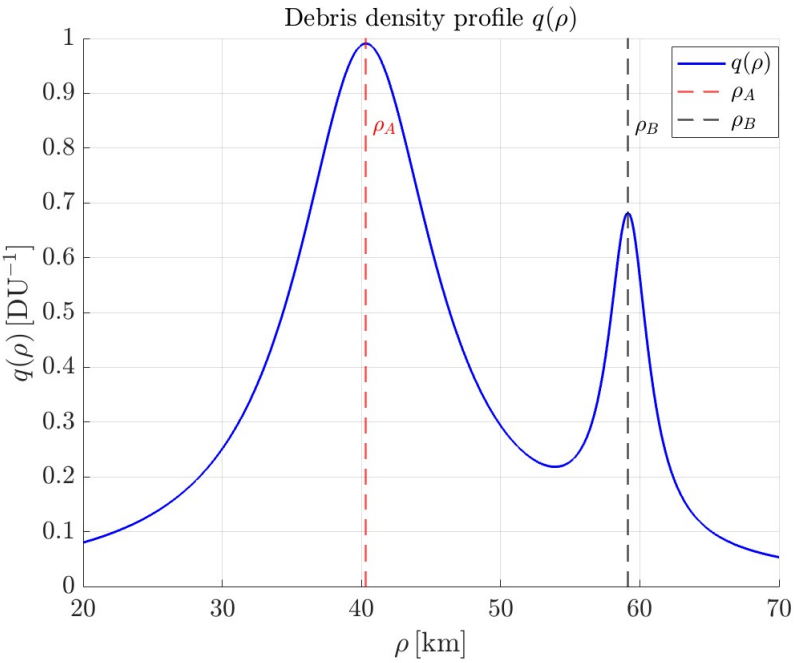


Figure 5: Debris density profile. Red and black dashed vertical lines mark Peak A and Peak B, respectively, where debris concentrations are highest.

– **Boundary State Vectors.** The dimensional initial and target state vectors in the ecliptic J2000 frame are reported in Table 9.

$r_{x,i}$ [km]	$r_{y,i}$ [km]	$r_{z,i}$ [km]	$v_{x,i}$ [km/s]	$v_{y,i}$ [km/s]	$v_{z,i}$ [km/s]
48.967144	48.967144	0.000000	−0.0015896	0.0015896	0.000000

$r_{x,f}$ [km]	$r_{y,f}$ [km]	$r_{z,f}$ [km]	$v_{x,f}$ [km/s]	$v_{y,f}$ [km/s]	$v_{z,f}$ [km/s]
53.150000	0.000000	0.000000	0.000000	0.002566	0.000000

Table 9: Initial and target state in Eros-centered ecliptic J2000 inertial frame.

The symmetry $x_0 = y_0$ and $|\dot{x}_0| = |\dot{y}_0|$ is consistent with a 45 placement on the circular orbit. The higher arrival velocity $v_f > v_i$ correctly reflects the energy gain required to maintain a lower circular orbit closer to Eros.

2.2 Adimensionalization Phase

- **Objective**
  - Convert all physical parameters from dimensional units (km, kg, s) into a consistent non-dimensional system to improve numerical conditioning of the TPBVP solver. The three base units DU, MU, TU are defined such that the scaled gravitational parameter satisfies $\bar{\mu} = 1$.
- **Methodology & Code Implementation**
  - **Reference Units.** The reference length and mass are prescribed directly:

$$DU = \rho_i = h_i + R_a = 69.25 \text{ km}, \qquad MU = m_0 = 25.0 \text{ kg} \tag{16}$$

The reference time TU is obtained by imposing $\bar{\mu} = 1$. From the dimensional conversion $\bar{\mu} = \mu\, TU^2/DU^3 = 1$:

$$TU = \sqrt{\frac{DU^3}{\mu}} = 3.0803 \times 10^4 \text{ s} \tag{17}$$

The derived velocity and force units follow as:

$$VU = \frac{DU}{TU}, \qquad FU = \frac{MU \cdot DU}{TU^2} \tag{18}$$

– **Parameter Conversion.** Each physical parameter is scaled by the appropriate combination of base units:

$$\begin{cases} \textbf{Thrust: } \bar{T} = \dfrac{T_{\max}\, TU^2}{MU \cdot DU \cdot 10^3} \quad \text{(converting N to km-based units before scaling)} \\ \textbf{Mass-flow rate: } \beta = \dfrac{\dot{m}_{\max}\, TU}{MU}, \quad \dot{m}_{\max} = \dfrac{T_{\max}}{I_{sp}\, g_0} \\ \textbf{Dust-peak radii: } \bar{\rho}_A = \dfrac{\rho_A}{DU}, \quad \bar{\rho}_B = \dfrac{\rho_B}{DU} \\ \textbf{State vectors: } \mathbf{r} = \dfrac{\mathbf{r}_{\text{dim}}}{DU}, \quad \mathbf{v} = \dfrac{\mathbf{v}_{\text{dim}}}{VU}, \quad m = \dfrac{m_{\text{dim}}}{m_0} \end{cases} \tag{19}$$

The density constants k_1, k_3 carry units [DU^{-1}] and k_2, k_4 carry [DU^2], so they are used directly in the non-dimensional cost function without further scaling.

- **Results & Analysis**

Table 10: Adimensionalized reference units and problem parameters.

Parameter	Symbol	Value	Unit
Reference length	DU	69.25	km
Reference time	TU	3.0803×10^4	s
Reference velocity	VU	2.2481×10^{-3}	km/s
Scaled grav. param.	$\bar{\mu}$	1.0	–
Scaled thrust	$\bar{T}$	1.1833×10^{-2}	–
Scaled mass-flow	β	7.0859×10^{-6}	–
Dust peak A	$\bar{\rho}_A$	5.8215×10^{-1}	–
Dust peak B	$\bar{\rho}_B$	8.5444×10^{-1}	–

The large values of $\bar{T}$ and β relative to the state components reflect the extremely low thrust-to-weight ratio of the electric propulsion system consistent with the multi-revolution transfer expected in Part 4. Setting $\bar{\mu} = 1$ ensures the gravitational term dominates the equations of motion with a unit coefficient, maximising numerical stability of the ODE integrator.

2.3 PMP Formulation and Zero-Finding Problem

- **Objective**

  – Apply Pontryagin's Minimum Principle (PMP) to derive the necessary conditions for the optimal low-thrust transfer, casting the problem as a Two-Point Boundary Value Problem (TPBVP) with unknowns $\{\boldsymbol{\lambda}_0, t_f\}$.

- **Methodology & Formulation**

– **State, Control, and Objective.** The augmented state vector is $\mathbf{x} = [\mathbf{r},\, \mathbf{v},\, m]^\top \in \mathbb{R}^7$. The control inputs are the throttle $u \in [0, 1]$ and the thrust-pointing unit vector $\hat{\boldsymbol{\alpha}}$, $\|\hat{\boldsymbol{\alpha}}\| = 1$. The cost functional to be minimised is

$$J = \int_{t_0}^{t_f} q(\rho(t))\,\mathrm{d}t, \qquad q(\rho) = \frac{k_1}{k_2 + \left(\dfrac{\rho - \rho_A}{DU}\right)^2} + \frac{k_3}{k_4 + \left(\dfrac{\rho - \rho_B}{DU}\right)^2} \tag{20}$$

where $\rho = \|\mathbf{r}\|$ and all quantities are expressed in the adimensional system of Part 2 ($\bar{\mu} = 1$, $DU = \rho_i$, $MU = m_0$).

– **Equations of Motion.** The spacecraft dynamics under Eros gravity and low-thrust propulsion are:

$$\dot{\mathbf{x}} = \begin{cases} \dot{\mathbf{r}} = \mathbf{v} \\ \dot{\mathbf{v}} = -\dfrac{\bar{\mu}}{\rho^3}\mathbf{r} + u\,\dfrac{\bar{T}}{m}\,\hat{\boldsymbol{\alpha}} \\ \dot{m} = -u\,\beta \end{cases} \tag{21}$$

where $\bar{T}$ is the adimensional maximum thrust and $\beta = \frac{\dot{m}_{\max} \cdot TU}{MU}$ is the adimensional mass-flow rate.

– **Hamiltonian and Costate Dynamics.** Introducing the costate vector $\boldsymbol{\lambda} = [\boldsymbol{\lambda}_r,\, \boldsymbol{\lambda}_v,\, \lambda_m]^\top$, the Hamiltonian and Euler-Lagrange costate equations are:

$$H = q(\rho) + \boldsymbol{\lambda}_r^\top \mathbf{v} - \frac{\bar{\mu}}{\rho^3}\boldsymbol{\lambda}_v^\top \mathbf{r} + u\,\frac{\bar{T}}{m}\,\boldsymbol{\lambda}_v^\top \hat{\boldsymbol{\alpha}} - u\,\beta\,\lambda_m \tag{22}$$

$$\Big\downarrow \text{costate dynamics: } \dot{\boldsymbol{\lambda}} = -\frac{\partial H}{\partial \mathbf{x}}$$

$$\dot{\boldsymbol{\lambda}} = \begin{cases} \dot{\boldsymbol{\lambda}}_r = -\nabla_{\mathbf{r}} q(\rho) - \dfrac{\bar{\mu}}{\rho^3}\boldsymbol{\lambda}_v + \dfrac{3\bar{\mu}}{\rho^5}(\boldsymbol{\lambda}_v^\top \mathbf{r})\,\mathbf{r} \\ \dot{\boldsymbol{\lambda}}_v = -\boldsymbol{\lambda}_r \\ \dot{\lambda}_m = -u\,\dfrac{\bar{T}}{m^2}\,\|\boldsymbol{\lambda}_v\| \end{cases} \tag{23}$$

where $\nabla_{\mathbf{r}} q = \frac{\mathrm{d}q}{\mathrm{d}\rho}\frac{\mathbf{r}}{\rho}$ and

$$\frac{\mathrm{d}q}{\mathrm{d}\rho} = \frac{-2k_1(\rho - \rho_A)}{[k_2 + (\rho - \rho_A)^2]^2} + \frac{-2k_3(\rho - \rho_B)}{[k_4 + (\rho - \rho_B)^2]^2} \tag{24}$$

– **Optimal Control Laws.** Minimising H over the admissible control set gives:

$$\begin{cases} \textbf{Thrust direction: } \hat{\boldsymbol{\alpha}}^* = -\dfrac{\boldsymbol{\lambda}_v}{\|\boldsymbol{\lambda}_v\|} \\ \textbf{Bang-bang throttle: } u^* = \begin{cases} 1 & \text{if } S < 0 \\ 0 & \text{if } S > 0 \end{cases}, \text{ with } S = -\dfrac{\bar{T}}{m}\|\boldsymbol{\lambda}_v\| - \beta\lambda_m \end{cases} \tag{25}$$

– **Boundary and Transversality Conditions.**

$$\begin{cases} \mathbf{r}(t_0) = \mathbf{r}_0, \quad \mathbf{v}(t_0) = \mathbf{v}_0, \quad m(t_0) = 1 & \text{(Initial state, Fixed)} \\ \mathbf{r}(t_f) = \mathbf{r}_f, \quad \mathbf{v}(t_f) = \mathbf{v}_f & \text{(Target state, Fixed)} \\ \lambda_m(t_f) = 0 & \text{(Free } m_f\text{, Transversality)} \\ H(t_f) = 0 & \text{(Free } t_f\text{, Transversality)} \end{cases} \tag{26}$$

Note: the freedom of m_f and t_f impose respectively $\lambda_m(t_f) = 0$ and $H(t_f)$=0.

– **Zero-Finding Problem.** The TPBVP is solved via single-shooting:

$$\text{Find } \mathbf{z} = [\boldsymbol{\lambda}_0; t_f] \in \mathbb{R}^8 \text{ s.t. } \mathbf{F}(\mathbf{z}) = \begin{bmatrix} \mathbf{r}(t_f) - \mathbf{r}_f \\ \mathbf{v}(t_f) - \mathbf{v}_f \\ \lambda_m(t_f) \\ H(t_f) \end{bmatrix} = \mathbf{0}_{8\times 1} \tag{27}$$

The Jacobian $\partial\mathbf{F}/\partial\mathbf{z}$ is approximated numerically via finite differences inside `fsolve` (Levenberg–Marquardt), avoiding the need to integrate the 14×14 variational equations and circumventing the Dirac-delta singularities introduced by the bang-bang switching law. The full augmented system integrates numerically the 14 coupled ODEs ($\mathbf{x} \in \mathbb{R}^7$, $\boldsymbol{\lambda} \in \mathbb{R}^7$) forward from t_0 to t_f.

2.4 Optimal Planar Transfer Solution ($\Delta i = 0$)

- **Objective** To numerically solve the TPBVP derived in Point 3, determining the optimal low-thrust trajectory that minimizes cumulative debris exposure during orbit lowering from $r_i = 69.25$ km to $r_f = 53.15$ km with $\Delta i = 0$.

- **Methodology & Code Implementation**

  – **Initial Guess Strategy.**
  Given the high sensitivity of the PMP shooting method to the initial costate guess, a stochastic search is performed by drawing shooting vectors with initial costates sampled from $\lambda_{r,0,i}, \lambda_{v,0,i} \in [-100, +100]$ and $\lambda_{m,0} \in [-100, +100]$, while the final time is sampled from $t_f \in [T_{\text{orb}}, 3T_{\text{orb}}]$ where $T_{\text{orb}} = 2\pi\sqrt{r_i^3/\bar{\mu}}$ is the initial circular orbital period;
  the lower bound excludes geometrically infeasible short transfers (the transfer from $\theta_i = 45°$ to $\theta_f = 0°$ corresponds to a 315° angular sweep, requiring at least one orbital period to complete considering prograde orbit), while the upper bound enforces the mission constraint of fewer than three revolutions.
  The best candidate, defined as the one minimizing the terminal state residual $e = \|[\mathbf{r}(t_f) - \mathbf{r}_f;\ \mathbf{v}(t_f) - \mathbf{v}_f]\|$ obtained by forward-integrating the augmented state–costate dynamics for each sampled vector, is then passed as the initial guess to `fsolve` solver.

  – **Shooting Method.** The TPBVP $\mathbf{F}(\mathbf{z}) = \mathbf{0}$ is solved via `fsolve` with the Levenberg–Marquardt algorithm, with the Jacobian approximated by finite differences and the augmented 14-ODE state-costate system numerically integrated. Once converged, this solution is reused for subsequent analyses without repeating the stochastic search.

  – **Hamiltonian Validation.** Since the system is autonomous with free t_f, PMP requires $H(t) = \text{const} = 0$ along the optimal arc. This was verified by computing $\Delta H(t) = H(t) - H(t_f)$ over the trajectory. The switching function $S(t)$ was monitored to confirm bang-bang thrust behavior.

- **Results & Analysis**

  The shooting solver converged with residual $\|\mathbf{F}\| = 6.3526 \times 10^{-12}$, confirming machine-precision satisfaction of boundary and transversality conditions. Tables 11 and 12 report the converged solution and terminal accuracy.

  The propellant consumption is $m_{\text{prop}} = 2.635$ g, corresponding to equivalent $\Delta v_{\text{eq}} = 0.3957$ m/s, confirming nearly ballistic motion with minimal thrust corrections.

Table 11: Optimal orbit lowering transfer solution ($\Delta i = 0$)

t_f [min]	m_f [kg]
7636.617583	24.997364

λ_{0,r_x}	λ_{0,r_y}	λ_{0,r_z}	λ_{0,v_x}	λ_{0,v_y}	λ_{0,v_z}	$\lambda_{0,m}$
−0.204453	−0.934755	0.000000	−0.010602	−0.649197	0.000000	2.251135

Table 12: Final state error with respect to target ($\Delta i = 0$)

Error	**Value**	**Units**
$\|\mathbf{r}(t_f) - \mathbf{r}_f\|$	7.428×10^{-8}	km
$\|\mathbf{v}(t_f) - \mathbf{v}_f\|$	3.476×10^{-9}	m/s

Figure 6 shows the optimal trajectory spiraling inward over approximately 2.4 revolutions from departure at +45 to arrival on the x-axis, remaining outside both debris peaks ρ_A and ρ_B for most of the arc.

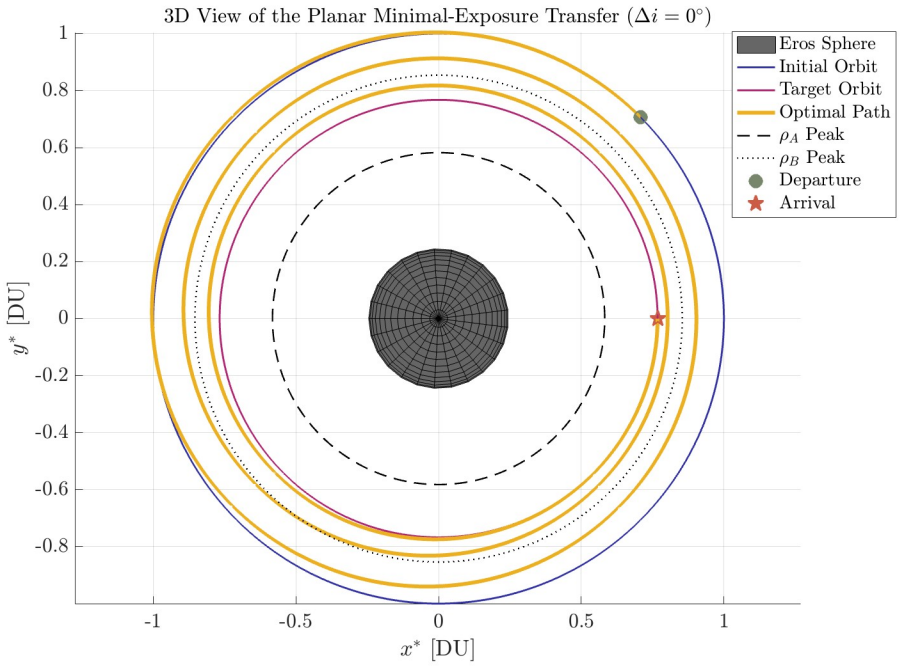


Figure 6: 3D view of the planar minimal-exposure transfer. The trajectory (orange) connects departure (green) to arrival (red), with dashed circles indicating debris peak radii.

Figure 7 confirms optimal control properties: Hamiltonian deviation ΔH remains bounded at $\mathcal{O}(10^{-10})$, with localized spikes near $t \approx 3$ days coinciding with sharp debris gradient crossings. The switching function $S(t) < 0$ (right) confirms full-thrust operation ($u^* = 1$) over the entire arc. Figure 8 shows radial distance decreasing monotonically from 69.25 km to 53.15 km (left), with debris exposure $q(\rho(t))$ peaking sharply near $t \approx 2.5$ days (crossing ρ_B) and exhibiting a secondary peak near $t \approx 4$ days, confirming that the optimizer minimizes time in hazard zones.

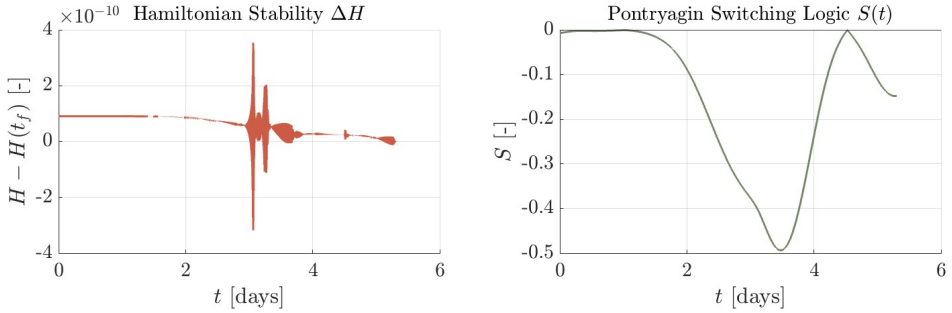


Figure 7: Hamiltonian stability ΔH (left) and switching function $S(t)$ (right) confirming optimality conditions.

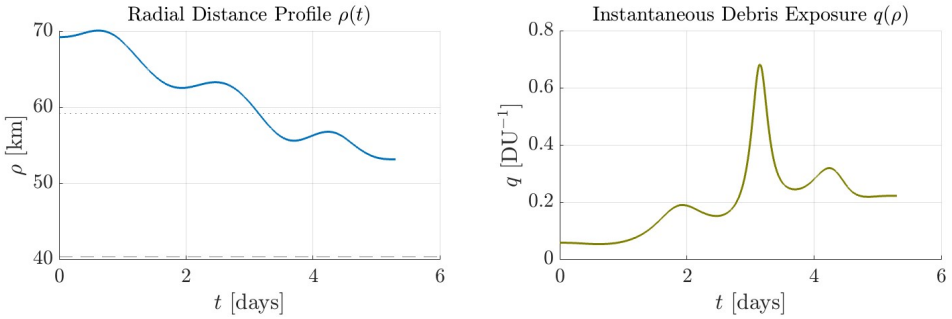


Figure 8: Radial distance $\rho(t)$ (left) and debris exposure $q(\rho)$ (right) throughout the transfer.

2.5 Numerical Continuation to Inclined Target Orbit ($\Delta i = 3.5$)

- **Objective**
  - To extend the planar optimal solution to a slightly inclined target orbit ($\Delta i = 3.5$, RAAN $= 0$, rotation about the x-axis of the ecliptic J2000 frame) using numerical continuation, and to compare the inclined solution with the planar one.
- **Methodology & Code Implementation**
  - **Continuation Strategy.** The continuation parameter is the inclination change Δi, discretized into 10 uniform increments from 0 to 3.5. At each iteration j, the target state is updated by rotating both position and velocity vectors through a direction-cosine matrix about the x-axis:

$$R_x(\Delta i_j) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\Delta i_j & -\sin\Delta i_j \\ 0 & \sin\Delta i_j & \cos\Delta i_j \end{bmatrix}, \qquad \mathbf{x}_f^{(j)} = \begin{bmatrix} R_x\,\mathbf{r}_f \\ R_x\,\mathbf{v}_f \end{bmatrix} \tag{28}$$

    Continuity across the continuation path is maintained by initializing each shooting solve with the costates and final time from the preceding converged step; the first step uses the planar solution as seed, so the solver always starts within the convergence region of the current target inclination.
  - **Shooting Solver.** At each continuation step, the shooting function is solved via Levenberg–Marquardt.The augmented 14-ODE state-costate system is numerically integrated.
  - **Performance Metrics.** After each solve, the achieved inclination is verified from the terminal angular momentum vector $\hat{\mathbf{h}} = \mathbf{r}(t_f) \times \mathbf{v}(t_f)$.

- **Results & Analysis**
  - **Convergence.** All 10 continuation steps converged with residuals ranging from 4.50×10^{-13} to 2.42×10^{-10}, confirming robust propagation of the solution across the inclination grid.
  - **Inclined Solution.** The converged unknowns at $\Delta i = 3.5$ are reported in Tables 13 and 14.

Table 13: Optimal transfer solution ($\Delta i = 3.5$).

t_f [min]	m_f [kg]
7568.311148	24.997388

λ_{0,r_x}	λ_{0,r_y}	λ_{0,r_z}	λ_{0,v_x}	λ_{0,v_y}	λ_{0,v_z}	$\lambda_{0,m}$
18.688876	15.762044	−35.839002	−18.763387	15.179919	−36.913936	10.691746

Table 14: Final state error with respect to inclined target ($\Delta i = 3.5$).

Error	**Value**	**Units**
$\|\mathbf{r}(t_f) - \mathbf{r}_f\|$	6.042×10^{-8}	km
$\|\mathbf{v}(t_f) - \mathbf{v}_f\|$	2.952×10^{-9}	m/s

  - **Comparison with Planar Solution.** Table 15 summarises the key differences between the two solutions.

Table 15: Planar vs. inclined solution comparison.

Metric	**Planar**	**Inclined**	**Units**
t_f^* [-]	14.875	14.742	–
t_f [days]	5.303	5.256	days
Δv_{eq}	0.3957	0.3922	m/s
$\max \lvert\Delta H\rvert$	3.54×10^{-10}	5.43×10^{-10}	–
Δi achieved	0.000	3.500	deg

The inclined solution yields a shorter transfer (t_f : 7636.62 → 7568.31 min) and a lower Δv_{eq} (0.3957 → 0.3922 m/s). The Δv_{eq} reduction is justified by the fact that propellant consumption scales directly with burn duration, which is identifiable only because $S(t) < 0$ throughout the arc and the bang-bang law therefore prescribes full thrust ($u^* = 1$) at every instant, thereby accommodating the out-of-plane plane change entirely by rotating the thrust pointing direction with no need to extend the arc.The Hamiltonian deviation remains $\mathcal{O}(10^{-10})$ for both cases.

  - **Optimality Validation:**

    For the maximum inclination case ($\Delta i = 3.5$), Figures 9–10 confirm optimality: $\Delta H \sim \mathcal{O}(10^{-10})$ verifies the transversality condition $H(t_f) = 0$, $S(t) < 0$ confirms full-thrust operation, and the radial/debris profiles (decreasing from 69.25 km to 53.15 km, peak near $t \approx 3$ days) remain qualitatively similar to the planar case.

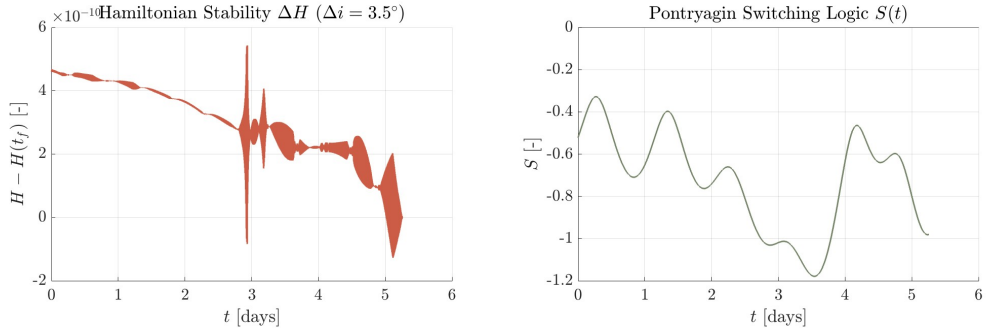


Figure 9: Inclined transfer: Hamiltonian stability (left) and switching function (right).

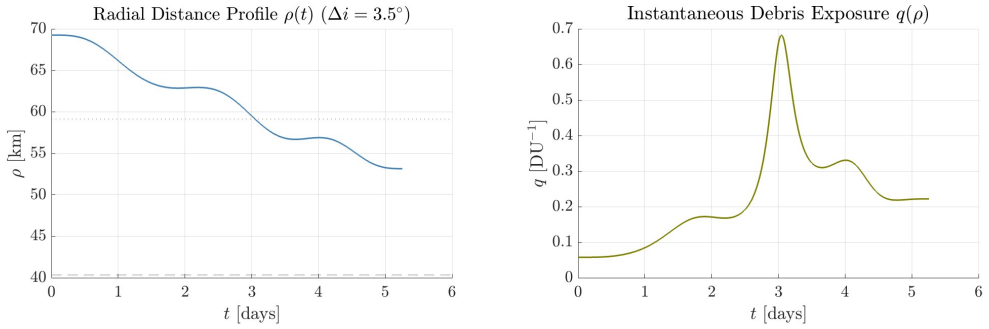


Figure 10: Inclined transfer: radial distance (left) and debris exposure (right).

– **Trajectory Visualization.** The 3D optimal trajectory (Figure 11) transitions from the ecliptic initial orbit to the inclined target orbit. The YZ-plane projection (Figure 12) clearly reveals the out-of-plane displacement resulting from the 3.5 inclination change.

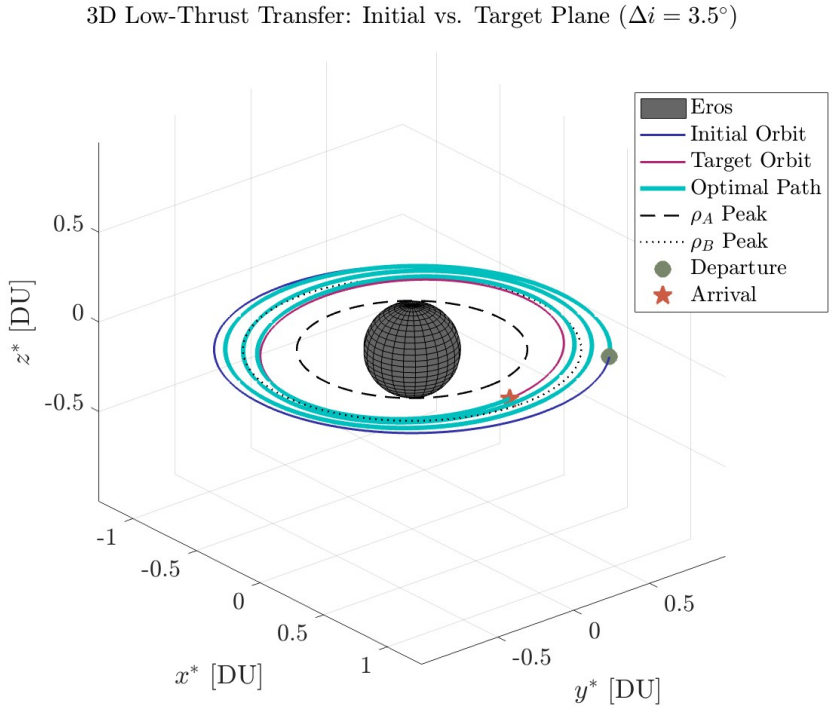


Figure 11: 3D view of the inclined minimal-exposure transfer ($\Delta i = 3.5$).

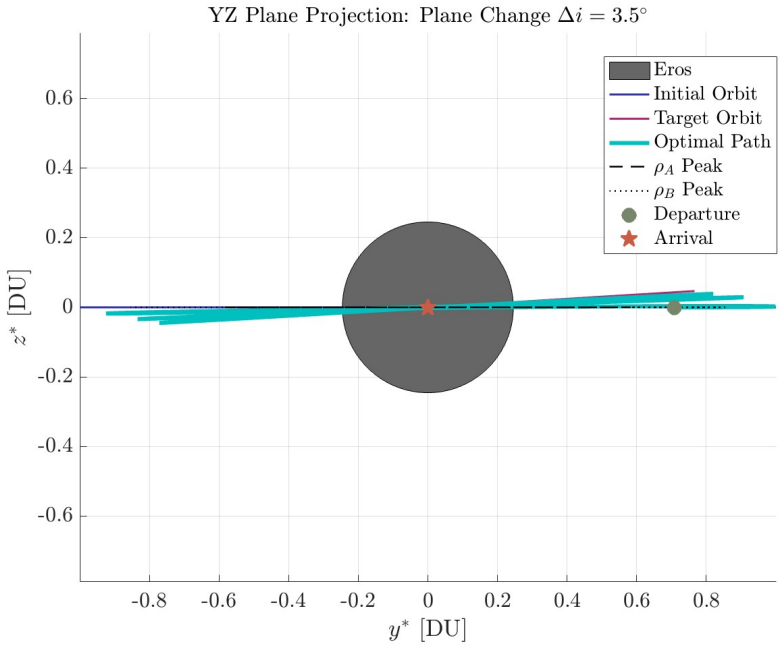


Figure 12: YZ-plane projection showing 3.5 plane change execution.

2.6 N-Body Propagation and Keplerian Comparison

- **Objective**
  - To assess the stability of the inclined optimal orbit obtained in Point 5 by propagating the terminal state under a high-fidelity n-body model and comparing the resulting trajectory against the ideal Keplerian (two-body) solution over a one-month interval.
- **Methodology & Code Implementation**
  - **Initial Condition.** The dimensional final state from the inclined optimal trajectory (Point 5) serves as the initial condition for N-body propagation:

$$\mathbf{x}_0^{\text{nb}} = \begin{bmatrix} \mathbf{r}(t_f) \cdot \text{DU} \\ \mathbf{v}(t_f) \cdot \text{VU} \end{bmatrix} \in \mathbb{R}^6 \quad [\text{km; km/s}] \tag{29}$$

    where the adimensional terminal state is converted to physical units using the normalization constants established in Point 1.
  - **N-Body Equations of Motion.** The spacecraft state evolves in the Eros-centered ECLIPJ2000 inertial frame under central Eros gravity plus third-body indirect perturbations from 11 solar-system bodies (Sun, Jupiter, Saturn, Earth, Moon, Mars, Venus, Uranus, Neptune, Mercury, Pluto). The equations of motion are:

$$\ddot{\mathbf{r}} = -\frac{\mu_{\text{Eros}}}{\|\mathbf{r}\|^3}\mathbf{r} + \sum_{i=1}^{11} \mu_i \left(\frac{\mathbf{d}_i^{\text{sc}}}{\|\mathbf{d}_i^{\text{sc}}\|^3} - \frac{\mathbf{d}_i^{\text{Eros}}}{\|\mathbf{d}_i^{\text{Eros}}\|^3} \right) \tag{30}$$

    where $\mathbf{d}_i^{\text{sc}} = \mathbf{r}_i - (\mathbf{r}_{\text{Eros}} + \mathbf{r})$ is the vector from perturbing body i to the spacecraft, and $\mathbf{d}_i^{\text{Eros}} = \mathbf{r}_i - \mathbf{r}_{\text{Eros}}$ is the vector from body i to Eros. The indirect term accounts for the non-inertial nature of the Eros-centered frame. Ephemeris data for all bodies are retrieved from SPICE kernels.

– **Keplerian Reference.** For comparison, the two-body Keplerian equations:

$$\ddot{\mathbf{r}} = -\frac{\mu_{\text{Eros}}}{\|\mathbf{r}\|^3}\mathbf{r} \tag{31}$$

– **Integration Settings.** Both N-body and Keplerian propagations are integrated from the same initial condition using `ode113` on an identical time grid spanning approximately 31 days from 2012-01-15T00:00:00 to 2012-02-15T00:00:00 TDB, enabling point-by-point trajectory comparison.

– **Error Decomposition.** To better characterize the expected divergence between the two models, the position difference $\Delta\mathbf{r} = \mathbf{r}_{\text{nb}} - \mathbf{r}_{\text{kep}}$ is decomposed into the local Radial-Transverse-Normal (RTN) frame of the Keplerian reference orbit:

$$\hat{\mathbf{R}} = \frac{\mathbf{r}_{\text{kep}}}{\|\mathbf{r}_{\text{kep}}\|}, \quad \hat{\mathbf{N}} = \frac{\mathbf{r}_{\text{kep}} \times \mathbf{v}_{\text{kep}}}{\|\mathbf{r}_{\text{kep}} \times \mathbf{v}_{\text{kep}}\|}, \quad \hat{\mathbf{T}} = \hat{\mathbf{N}} \times \hat{\mathbf{R}} \tag{32}$$

This RTN decomposition separates the position error into radial, transverse, and normal components, isolating gravitational perturbation effects along each orbital direction.

- **Results & Analysis**

Figure 13 shows visually indistinguishable N-body and Keplerian trajectories over 31 days, confirming Eros gravity dominance. Table 16 quantifies the divergence: final position error reaches 0.161 km with RMS of 0.0932 km.

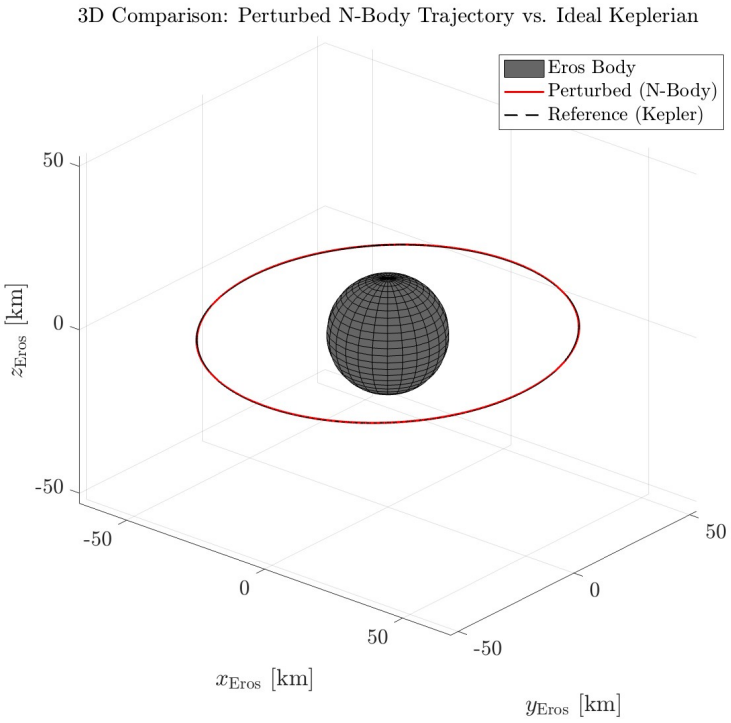


Figure 13: N-body (red) versus Keplerian (dashed) trajectories: visually coincident over 31-day propagation.

Table 16: Position error statistics for $\Delta\mathbf{r} = \mathbf{r}_{\text{nbody}} - \mathbf{r}_{\text{kep}}$: N-body vs. Keplerian propagation

Metric	Value	Units
Final $\|\Delta\mathbf{r}\|$	0.1609	km
Max $\|\Delta\mathbf{r}\|$	0.1609	km
RMS $\|\Delta\mathbf{r}\|$	0.0932	km
Max Radial (dR)	2.87×10^{-3}	km
Max Transverse (dT)	1.61×10^{-1}	km
Max Normal (dN)	8.62×10^{-3}	km
RMS Radial	1.31×10^{-3}	km
RMS Transverse	9.31×10^{-2}	km
RMS Normal	3.24×10^{-3}	km

The RTN decomposition (Figure 14) reveals transverse-dominated divergence: dT reaches 0.161 km while radial and normal components remain below 10 meters. This behavior is characteristic of secular along-track drift induced by third-body perturbations causing small orbital period shifts. The two-body model suffices for preliminary trajectory design over short arcs, while the N-body formulation becomes essential for high-precision applications such as long-duration propagation.

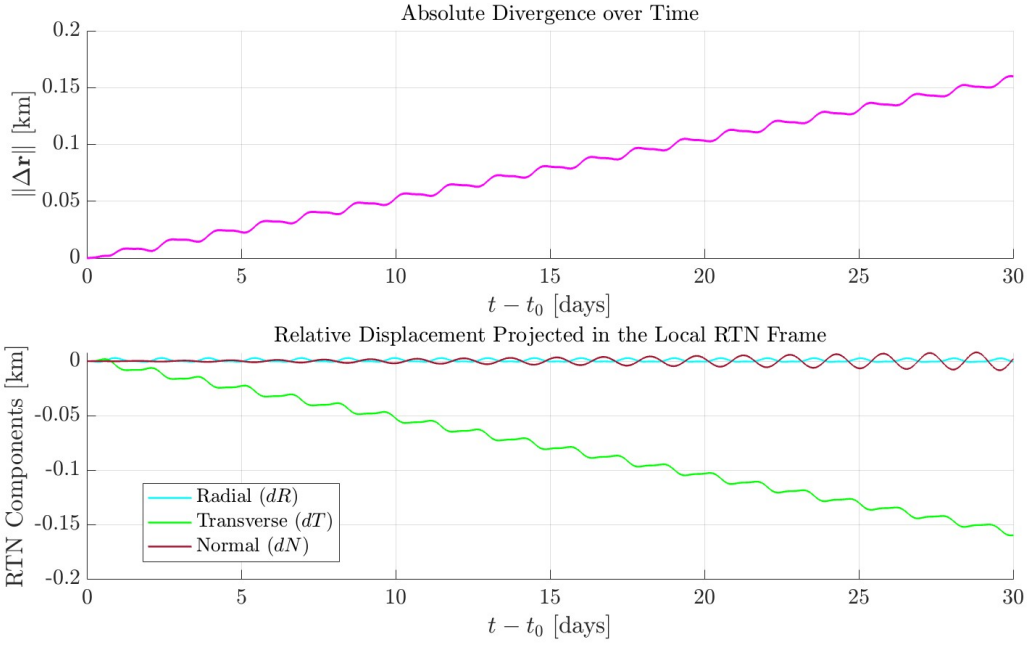


Figure 14: Position error growth (top) and RTN decomposition (bottom) showing transverse-dominated secular drift.

3 Impulsive Guidance

Exercise 3

The LUMOS mission has the objective of enabling cislunar space surveillance, which is crucial to ensure space safety for both current and future missions. Equipped with an optical payload, it will observe and monitor a region of interest around the Moon to identify and track the motion of objects within this domain.

Among the candidate orbits for this mission, a Distant Prograde Orbit (DPO) ensures good performance in terms of relative visibility between LUMOS and the expected population of objects around the Moon. The DPO has therefore been selected as the operational orbit for the LUMOS mission. The associated orbit kernel is provided.

You have been asked to compute the transfer trajectory from an Earth orbit to the target DPO which minimizes the fuel consumptions using at least two impulsive maneuvers, one placed at the beginning and one at the end of the transfer. Intermediate maneuvers can be placed at the control nodes of a shooting method. The initial conditions are provided at the initial time t_0, which correspond to a state on the nominal injection trajectory, after separation from the launcher.

x [km]	y [km]	z [km]
4882.128777349525	4030.951188974080	1444.063996364637

v_x [km/s]	v_y [km/s]	v_x [km/s]
−5.761404259200368	9.117460162775972	−2.399524966790392

Table 17: Initial state $\mathbf{x}_0 = \mathbf{x}(t_0)$ in Earth-centered equatorial J2000 inertial frame.

1) Using the initial conditions provided in Table 18, perform an n-body propagation taking the Earth as the central body. Consider a propagation interval from the initial epoch $t_0 =$ `2032-11-14T05:16:23.639` to the epoch $t_A =$ `2033-02-14T07:45:05.552`. Plot the obtained solution and report the error (position and velocity components) at time t_A with respect to the target orbit.

2) Formalize an unambiguous statement of the problem for a multiple shooting method specifying the considered optimization variables, objective function, any linear and non-linear equality and inequality constraints, considering that:

   - the initial epoch t_0 is fixed, together with the initial position $\mathbf{r}_0 = \mathbf{r}(t_0)$
   - an impulsive maneuver at t_0 is required to insert the spacecraft on the transfer orbit
   - the final target position $\overline{\mathbf{r}}_f = \overline{\mathbf{r}}(t_f)$ is on the target DPO.
   - a final impulsive maneuver at t_f is needed to acquire the target DPO
   - the final time t_f is free.

3) Considering the first guess solution given in Point 1, solve the transfer problem with multiple shooting taking $N = 5$. Provide and discuss the optimization results, plot the obtained transfer, and report the error on the final position at t_f.

(11 points)

3.1 N-Body Propagation and DPO Miss Analysis

- **Objective** To propagate the spacecraft state from the initial epoch t_0 (2032-11-14T05:16:23 UTC) to the arrival epoch t_A (2033-02-14T07:45:05 UTC) using an N-body dynamical model centred on Earth in the J2000 inertial reference frame, and to evaluate the miss distance with respect to the Distant Prograde Orbit (DPO) ephemeris at the nominal arrival time.

- **Methodology & Code Implementation** The propagation employed a high-fidelity N-body model accounting for gravitational perturbations from multiple Solar System bodies. The implementation follows these steps:

  1. **Dynamical Model Selection:** The spacecraft motion was governed by the Earth-centred equations of motion with third-body perturbations from 10 celestial bodies: Sun, Moon, Mercury, Venus, Mars, Jupiter, Saturn, Uranus, Neptune, and Pluto. The acceleration formulation uses the Encke q-method to minimize numerical errors in third-body terms

     Position queries were performed at each integration step to ensure high-accuracy perturbation modeling.
  2. **Propagation Setup.** The spacecraft is initialized at t_0 in the Earth-centred J2000 frame with $\mathbf{r}_0 = [4882.129,\ 4030.951,\ 1444.064]^T$ km and $\mathbf{v}_0 = [-5.761,\ 9.117,\ -2.400]^T$ km/s. All planetary positions and the DPO ephemeris are retrieved from SPICE kernels at each integration step, and the equations of motion are integrated via `ode113` with a constrained maximum step size to ensure accurate resolution of perturbation variations throughout the transfer arc.
  3. **Terminal State Comparison:** At the arrival epoch t_A, the propagated state $\mathbf{x}(t_A)$ was compared against the DPO ephemeris state retrieved from SPICE to compute position and velocity errors.

- **Results & Analysis** The uncontrolled propagation from t_0 to t_A resulted in significant deviation from the DPO target orbit. The miss distance and velocity error are quantified in Table 18.

Table 18: State vector error analysis at nominal arrival epoch t_A

Metric	Value
Position Error x [km]	21985.530452
Position Error y [km]	23789.843065
Position Error z [km]	10651.525243
Velocity Error x [km/s]	-0.110812
Velocity Error y [km/s]	0.194590
Velocity Error z [km/s]	0.055526
$\|\mathbf{r}(t_A) - \mathbf{r}_{\text{DPO}}\|$ [km]	34144.181472
$\|\mathbf{v}(t_A) - \mathbf{v}_{\text{DPO}}\|$ [m/s]	230.768406

The position miss distance of 34144.18 km and velocity error of 230.768406. m/s demonstrate that purely ballistic flight (without corrective maneuvers) is insufficient to achieve DPO rendezvous. The substantial magnitude of these errors motivates the need for an optimized impulsive transfer strategy, which will be addressed in Point 3.

Figure 15 illustrates the propagated trajectory in the Earth-centred J2000 reference frame. The spacecraft follows a highly elliptical orbit influenced by lunar and solar perturbations, but fails to intercept the DPO target state (blue marker) at the nominal arrival time t_A.

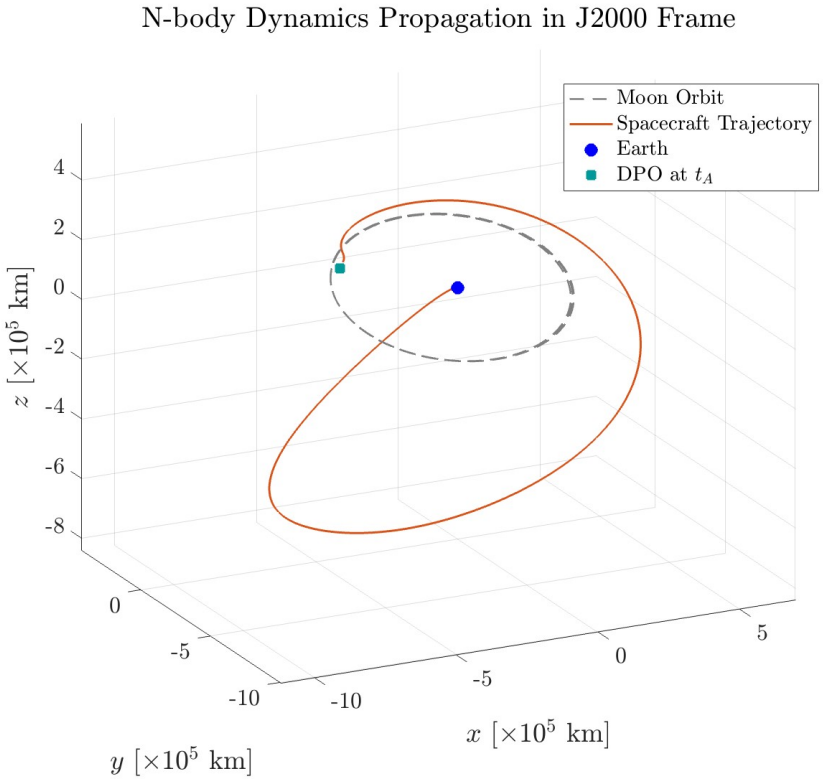


Figure 15: N-body dynamics propagation in J2000 frame.

3.2 Optimal Control Problem Formulation

- **Objective** To formally state the multiple-shooting impulsive transfer optimization problem with free final time, defining the decision variables, constraints, and objective function that will guide the numerical solution in Point 3.

- **Problem Formulation** The transfer from the initial Earth orbit to the target DPO is cast as a constrained nonlinear programming problem and solved via a multiple-shooting approach. The trajectory is decomposed into $N = 5$ nodes (yielding 4 segments) with impulsive velocity changes $\Delta\mathbf{v}_i$ applied at each node $i \in \{0, 1, 2, 3, 4\}$. The initial epoch t_0 is fixed together with $\mathbf{r}_0 = \mathbf{r}(t_0)$, while the final epoch is free and parameterized as $t_f = t_0 + \Delta T$. Node epochs are uniformly spaced over the transfer interval:

$$t_k = t_0 + k\,h, \qquad h = \frac{\Delta T}{N-1}, \qquad k = 0, \ldots, N-1. \tag{33}$$

  – **Decision Vector:** The optimization variables are collected into a 28-dimensional vector:

$$\mathbf{X} = [\Delta\mathbf{v}_0^T,\ (\mathbf{r}_1,\ \Delta\mathbf{v}_1)^T, (\ \mathbf{r}_2,\ \Delta\mathbf{v}_2)^T, (\ \mathbf{r}_3,\ \Delta\mathbf{v}_3)^T, (\ \mathbf{r}_f,\ \Delta\mathbf{v}_4)^T,\ \Delta T]^T \in \mathbb{R}^{28} \tag{34}$$

  comprising:

$$\begin{cases} \Delta\mathbf{v}_i \in \mathbb{R}^3 \text{ for } i = 0, \ldots, 4 \text{ (15 variables): impulsive velocity increments} \\ \mathbf{r}_j \in \mathbb{R}^3 \text{ for } j = 1, 2, 3, f \text{ (12 variables): intermediate and final position nodes} \\ \Delta T \in \mathbb{R}^+ \text{ (1 variable): total time of flight from } t_0 \text{ to } t_f \end{cases} \tag{35}$$

– **Objective Function:**
The natural performance index $\Delta V = \sum_{i=0}^{4} \|\Delta \mathbf{v}_i\|$ is non-smooth at any zero impulse, which degrades the local quadratic model used by `fmincon` at iterations where some increments are small. A squared expression is therefore minimized instead:

$$J(\mathbf{X}) = \sum_{i=0}^{4} \|\Delta \mathbf{v}_i\|^2, \tag{36}$$

which is everywhere differentiable and yields a well-conditioned Hessian approximation throughout the SQP iterations. The physical ΔV budget is recovered a posteriori as $\Delta V = \sum_{i=0}^{4} \|\Delta \mathbf{v}_i\|$ once convergence is achieved.

– **Equality Constraints:** The trajectory must satisfy segment continuity (defect constraints) and terminal matching conditions. Let $\boldsymbol{\varphi}(\mathbf{x}_j, t_j, t_{j+1})$ denote the flow map propagating the post-impulse state $\mathbf{x}_j = [\mathbf{r}_j;\, \mathbf{v}_j^- + \Delta \mathbf{v}_j] = [\mathbf{r}_j;\, \mathbf{v}_j^+]$ over $[t_j, t_{j+1}]$ under N-body dynamics, with $\boldsymbol{\varphi}_{\mathbf{r}}$ and $\boldsymbol{\varphi}_{\mathbf{v}}$ denoting its position and velocity components respectively.

1. **Segment Defects (12 constraints):** Position continuity at internal nodes:

$$\begin{cases} \mathbf{z}_1 := \boldsymbol{\varphi}_{\mathbf{r}}([\mathbf{r}_0,\, \mathbf{v}_0 + \Delta \mathbf{v}_0],\, t_0, t_1) - \mathbf{r}_1 = \mathbf{0} \\ \mathbf{z}_2 := \boldsymbol{\varphi}_{\mathbf{r}}([\mathbf{r}_1,\, \boldsymbol{\varphi}_{\mathbf{v}}^{(1)} + \Delta \mathbf{v}_1],\, t_1, t_2) - \mathbf{r}_2 = \mathbf{0} \\ \mathbf{z}_3 := \boldsymbol{\varphi}_{\mathbf{r}}([\mathbf{r}_2,\, \boldsymbol{\varphi}_{\mathbf{v}}^{(2)} + \Delta \mathbf{v}_2],\, t_2, t_3) - \mathbf{r}_3 = \mathbf{0} \\ \mathbf{z}_4 := \boldsymbol{\varphi}_{\mathbf{r}}([\mathbf{r}_3,\, \boldsymbol{\varphi}_{\mathbf{v}}^{(3)} + \Delta \mathbf{v}_3],\, t_3, t_f) - \mathbf{r}_f = \mathbf{0} \end{cases} \tag{37}$$

where $\boldsymbol{\varphi}_{\mathbf{v}}^{(j)}$ is the arrival velocity at node j propagated from the preceding segment.

2. **Terminal Rendezvous (6 constraints):** Position and velocity matching with the DPO ephemeris at the free final time $t_f = t_0 + \Delta T$:

$$\begin{cases} \boldsymbol{\psi}_f := \mathbf{r}_f - \mathbf{r}_{\mathrm{DPO}}(t_f) = \mathbf{0} \\ \boldsymbol{\psi}_v := \boldsymbol{\varphi}_{\mathbf{v}}^{(4)} + \Delta \mathbf{v}_4 - \mathbf{v}_{\mathrm{DPO}}(t_f) = \mathbf{0} \end{cases} \tag{38}$$

– **Bound Constraints:** The time of flight is restricted to a physically reasonable range:

$$\Delta T_{\min} \le \Delta T \le \Delta T_{\max} \tag{39}$$

where $\Delta T_{\min}$ and $\Delta T_{\max}$ define the admissible transfer duration window, preventing unrealistically short or long solutions and improving numerical robustness.

– **Consolidated Problem Statement:**

$$\begin{cases} \min_{\mathbf{X}} \quad J(\mathbf{X}) = \sum_{i=0}^{4} \|\Delta \mathbf{v}_i\|^2 \\ \text{subject to} \quad \begin{cases} \mathbf{c}_{\mathrm{eq}}(\mathbf{X}) = \mathbf{0} \in \mathbb{R}^{18} \\ \Delta T_{\min} \le \Delta T \le \Delta T_{\max} \end{cases} \end{cases} \tag{40}$$

This nonlinear programming problem is solved numerically via MATLAB's `fmincon` in Point 3.

3.3 Multiple-Shooting Optimization Solution

- **Objective** To solve the formulated optimal control problem via sequential quadratic programming, determining the optimal impulse schedule, trajectory node positions, and transfer time that minimize total ΔV while achieving precise DPO rendezvous.

- **Methodology & Code Implementation** The optimization was executed using MATLAB's `fmincon` with the following implementation strategy:

  1. **Initial Guess Generation:**
     - A warm-start strategy was employed. The time grid was uniformly discretized as $t_i = t_0 + i \cdot \Delta T_{\text{nom}}/(N-1)$for $i = 0, \ldots, 4$.
     - Intermediate node positions $\mathbf{r}_j$ were obtained by sequentially propagating ballistic arcs under N-body dynamics from t_0 to each node epoch.
     - All impulse guesses were initialized to zero ($\Delta\mathbf{v}_i^{\text{guess}} = \mathbf{0}$), since the unforced propagation already provides a geometrically consistent trajectory with negligible velocity discontinuities at the nodes. The nominal time of flight $\Delta T_{\text{nom}} = t_A - t_0 \approx 91.2$ days was used as the initial ToF guess.
  2. **Constraint Function Implementation:**
     - At each solver iteration, the equality constraint function extracts the decision variables, builds the current time grid from ΔT, and propagates each segment via `ode113` under the same N-body model as Point 1. Position defects are evaluated by comparing propagated arrival positions with the node variables.
     - The DPO ephemeris is queried at the free final time t_f via SPICE, and the terminal constraint compares the post-impulse final state with the target.
     - All position constraints are normalized by a fixed scaling factor `position_scaling` $= 10^3$ km, chosen to balance the order-of-magnitude difference between position residuals ($\sim 10^4$ km) and velocity residuals ($\sim 10^{-1}$ km/s), improving the numerical conditioning of the constraint Jacobian seen by solver.

- **Results & Analysis**

  The optimizer converged in 31 iterations with a maximum constraint violation of 9.148×10^{-9} and `exitflag` $= 1$, confirming satisfaction of all equality constraints to within numerical precision. The optimized transfer parameters are summarized in Table 19

Table 19: Optimized transfer summary

Metric	Value
Optimal Arrival Time t_f [UTC]	2033-02-14T03:31:25.273
Time of Flight ΔT [days]	91.9271
Total ΔV [m/s]	108.4812
$\|\mathbf{r}(t_f) - \mathbf{r}_{\text{DPO}}(t_f)\|$ [km]	1.6504×10^{-11}
$\|\mathbf{v}(t_f) - \mathbf{v}_{\text{DPO}}(t_f)\|$ [m/s]	7.457×10^{-8}

The terminal errors are negligible, confirming precise DPO insertion. The optimized arrival occurs approximately 4.23 hours earlier than the nominal epoch, with the free final time allowing the optimizer to reduce the impulse cost while enforcing the terminal boundary conditions.

The optimized impulse schedule is reported in Table 20. The impulse budget is heavily concentrated at the terminal node: $\|\Delta\mathbf{v}_4\| = 81.90$ m/s represents approximately three quarters of the total ΔV, as expected for a rendezvous problem where the final velocity acquisition dominates. The three preceding small impulses collectively steer the trajectory geometry, $\Delta\mathbf{v}_3$ applies a late-arc correction, and $\Delta\mathbf{v}_4$ enforces the full state match at t_f.

The post-insertion velocity is reported in Table 21.

Figure 16 shows the optimized trajectory with the five control nodes and four connecting arcs.

Table 20: Optimized impulse schedule [m/s]

Epoch [UTC]	Δv_x	Δv_y	Δv_z	$\|\Delta \mathbf{v}\|$
2032-11-14T05:16:23.638	−2.562651	−1.613427	−0.061946	3.028888
2032-12-07T04:50:07.753	+1.999842	+0.333837	+0.252235	2.043144
2032-12-30T04:23:51.867	+1.545690	+1.101053	+0.490785	1.960190
2033-01-22T03:57:35.981	−0.217661	−15.866184	−11.856771	19.808236
2033-02-14T03:31:25.273	−81.261156	−7.815559	+0.863693	81.640703

Table 21: Post-insertion velocity in Earth-centred J2000 frame

v_x [km/s]	v_y [km/s]	v_z [km/s]
−5.763966	9.115847	−2.399586

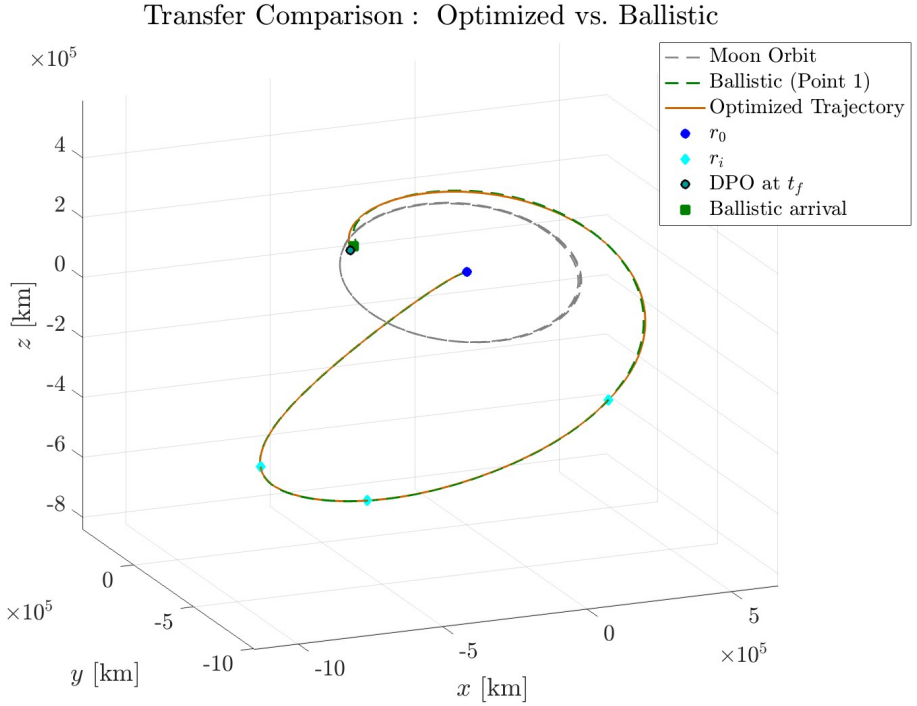


Figure 16: Optimized and Non-Optimized transfer in Earth-centred J2000 frame. Control nodes: $\mathbf{r}_0$ (blue), intermediate nodes (cyan), DPO at t_f (purple).

- **Alternative Formulation** An alternative multiple-shooting solution was computed with $N = 5$ segments and $N + 1 = 6$ impulsive maneuvers, adopting a full-state node decision vector:

$$\mathbf{X} = [\mathbf{v}_0^{+T},\ (\mathbf{r}_1, \mathbf{v}_1)^T,\ (\mathbf{r}_2, \mathbf{v}_2)^T,\ (\mathbf{r}_3, \mathbf{v}_3)^T,\ (\mathbf{r}_4, \mathbf{v}_4)^T,\ \Delta T]^T \in \mathbb{R}^{28} \tag{41}$$

comprising:

$$\begin{cases} \mathbf{v}_0^+ = \mathbf{v}_0 + \Delta \mathbf{v}_0 \in \mathbb{R}^3 \text{ (3 variables): post-impulse velocity at } t_0 \\ (\mathbf{r}_k, \mathbf{v}_k) \in \mathbb{R}^6 \text{ for } k = 1, 2, 3, 4 \text{ (24 variables): full post-impulse state at each node} \\ \Delta T \in \mathbb{R}^+ \text{ (1 variable): total time of flight from } t_0 \text{ to } t_f \end{cases} \tag{42}$$

Node epochs are uniformly spaced as:

$$t_k = t_0 + k\,h, \qquad h = \frac{\Delta T}{N}, \qquad k = 0, \ldots, N \tag{43}$$

The impulse at each intermediate node is implicitly defined as the velocity jump:

$$\Delta\mathbf{v}_k = \mathbf{v}_k - \boldsymbol{\varphi}_{\mathbf{v}}([\mathbf{r}_{k-1};\, \mathbf{v}_{k-1}^+],\, t_{k-1}, t_k), \qquad k = 1, \ldots, 4 \tag{44}$$

where $\mathbf{v}_k$ is the post-impulse velocity at node k (a decision variable) and $\boldsymbol{\varphi}_{\mathbf{v}}(\cdot, t_{k-1}, t_k)$ is the velocity component of the flow map propagated from the preceding arc. A final acquisition impulse closes the rendezvous:

$$\Delta\mathbf{v}_5 = \mathbf{v}_{\mathrm{DPO}}(t_f) - \boldsymbol{\varphi}_{\mathbf{v}}([\mathbf{r}_4;\, \mathbf{v}_4^+],\, t_4, t_f) \tag{45}$$

The squared total ΔV is minimized for differentiability:

$$J(\mathbf{X}) = \sum_{i=0}^{5} \|\Delta\mathbf{v}_i\|^2 \tag{46}$$

Only position defects and terminal position matching are enforced as hard equality constraints (15 constraints vs. 18 in the primary formulation), with the terminal velocity match $\boldsymbol{\psi}_v$ removed and absorbed into the cost via $\Delta\mathbf{v}_5$:

$$\begin{cases} \mathbf{z}_k := \boldsymbol{\varphi}_{\mathbf{r}}([\mathbf{r}_{k-1};\, \mathbf{v}_{k-1}^+],\, t_{k-1}, t_k) - \mathbf{r}_k = \mathbf{0}, \quad k = 1, \ldots, 4 \quad \text{(12 defects)} \\ \boldsymbol{\psi}_f := \boldsymbol{\varphi}_{\mathbf{r}}([\mathbf{r}_4;\, \mathbf{v}_4^+],\, t_4, t_f) - \mathbf{r}_{\mathrm{DPO}}(t_f) = \mathbf{0} \quad \text{(3 terminal position)} \end{cases} \tag{47}$$

Position residuals are normalized by `position_scaling`$= 10^3$ km for numerical conditioning. The time of flight is also bounded to allow admissible transfer duration window as in Section 3.2: The key motivation is that removing the hard terminal velocity constraint $\boldsymbol{\psi}_v$ reduces the active constraint count at fixed problem dimension, enlarging the feasible domain and giving the solver greater freedom to find lower-cost solutions.

The optimizer converged in 35 iterations (`exitflag` $= 1$, $\max|\mathbf{c}_{\mathrm{eq}}| = 2.147 \times 10^{-8}$), compared to 31 for the primary formulation. The optimized arrival epoch is 2033-02-14T07:54:13.063 UTC. The solution achieves $\Delta V = 96.46$ m/s, a **11.07%** reduction over the previous formulation result of 108.46 m/s, confirming the theoretical argument. All six impulses are active, representing a true multi-impulse transfer with magnitudes $\|\Delta\mathbf{v}_0\| = 0.27$ m/s, $\|\Delta\mathbf{v}_1\| = 0.07$ m/s, $\|\Delta\mathbf{v}_2\| = 0.38$ m/s, $\|\Delta\mathbf{v}_3\| = 5.57$ m/s, $\|\Delta\mathbf{v}_4\| = 7.44$ m/s, and $\|\Delta\mathbf{v}_5\| = 82.74$ m/s.

The terminal position error $\|\mathbf{r}(t_f) - \mathbf{r}_{\mathrm{DPO}}\| = 2.48 \times 10^{-5}$ km is larger than the previous formulation result of 1.65×10^{-11} km, a direct consequence of the relaxed constraint set. Terminal velocity error is zero by construction via $\Delta\mathbf{v}_5$.

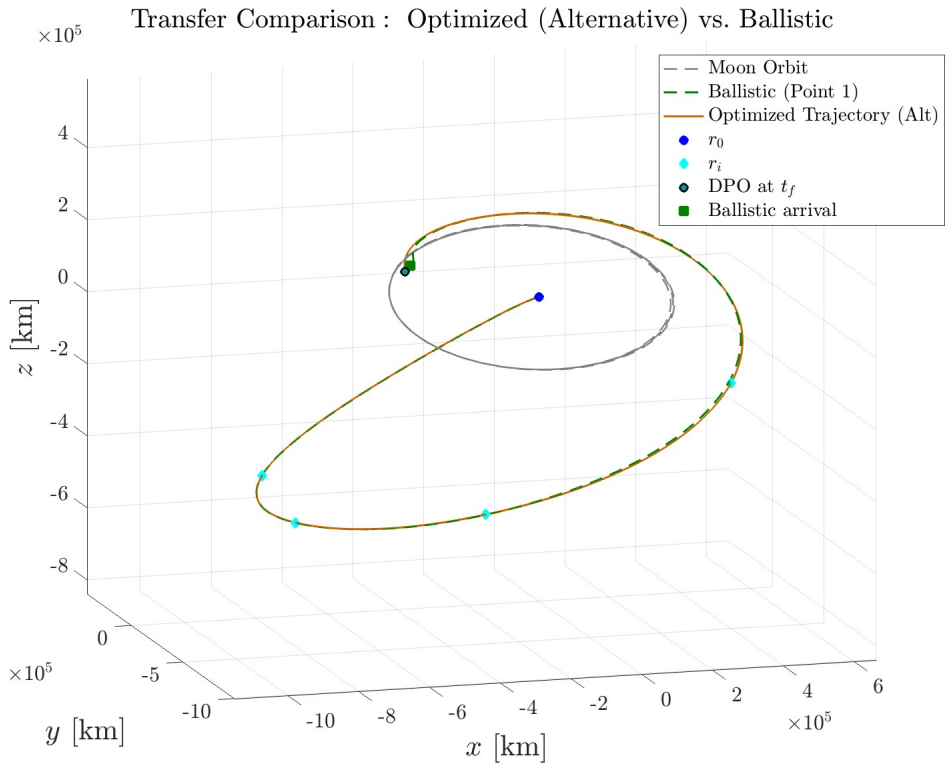


Figure 17: Alternative formulation: optimized transfer trajectory in Earth-centred J2000 frame. Control nodes: $\mathbf{r}_0$ (blue), intermediate nodes (cyan), DPO at t_f (teal).